engineering 4 us all

Student Handout

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MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - Bollards

“How Bollards Enhance Bike Lanes and Bike Parking: 1-800-BOLLARDS.” 1, 12 Sept. 2024, www.1800bollards.com/how-bollards-enhance-bike-lanes-and-bike-parking/?srsltid=AfmBOopx7bzPS7LUBF2YjMP0gf-4az1RP4EhiGDINs_CFX5eAn-USzKD.

With bike lanes being close to vehicle traffic, bollards are small barrier devices that divide the road from the bike lane. They can prevent oncoming traffic from making contact with bikers, and can be a mental reassurance to bikers that they are safer and therefore can focus on the road more. Bollards can also act as illumination devices if there is no natural light for bikers to see the road

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This method is successful; however, it does not solve the problem completely because they are not implemented everywhere. It fails to solve the issue of bike deaths due to the fact that it is not widely used everywhere.

Prior Design Attempt #2 - Annapolis - West Street Separated Bike Lane (2024)

https://www.annapolis.gov/212/Annapolis-Area-Bike-Map

As a busy city in Maryland, Annapolis, though pedestrian friendly, is not very bike friendly, especially on busy streets like West Street. To help fix this problem, the city installed a separated bike lane in 2024, giving bikers safer access to a busy and eventful street. The bike lanes use reflective posts as well as make shift curbs to protect bikers. These makeshift protections are enough for Annapolis because the tight roads force drivers to go slowly meaning the bikers don't need as much protection.

These bike lanes are very efficient due to the limited space, making it harder for both bikers and drivers to navigate. Annapolis streets are already tight enough as the city is very old

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and wasn’t originally designed for cars. The addition of these bike lanes in this new city are very controversial for this reason.

Though this attempt was successful in giving bikers a protected lane, it is unsuccessful in the way that it causes a nuisance to the drivers who are already forced to navigate tight and crowded streets.

Prior Design Attempt #3 - <NACTO Urban Bikeway Design Guide>

National Association of City Transportation Officials (NACTO). “Designing Protected Bike Lanes.” NACTO Urban Bikeway Design Guide.

Protected bike lane barriers are physical dividers placed between bike lanes and car lanes, often made from concrete curbs, planters, or plastic posts. These designs are meant to give cyclists a greater sense of safety by creating a strong visual and physical separation from vehicle traffic. Many cities have started using these barriers to encourage more people to bike by reducing the risk of collisions and improving overall safety for both cyclists and drivers.

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While protected bike lane barriers are effective in increasing cyclist safety and visibility, they are expensive to install and maintain compared to simpler methods like painted lines or bollards. They also take up more road space, which can lead to traffic congestion in smaller streets. In some cases, poor maintenance or unclear markings cause confusion for both cyclists and drivers. Overall, this design helps solve many safety issues but is not always practical or affordable for all cities to implement widely.

Prior Design Attempt #4 - <U.S. Department of Transportation, Federal Highway Administration>

U.S. Department of Transportation, Federal Highway Administration (FHWA). “Bikeway Design Best Practices.” FHWA Safety Program.

Painted bike lanes are one of the earliest and most common designs for separating bicycles from motor vehicles. These lanes are created using colored pavement or painted lines on existing roads to mark a clear space for cyclists. They are inexpensive, easy to install, and help drivers become more aware of cyclists on the road. Many cities began using painted lanes as a quick way to promote bike commuting without major construction.

While painted bike lanes are low-cost and simple to add, they do not offer real physical protection from cars. Vehicles can easily drift into the lanes, and drivers sometimes ignore

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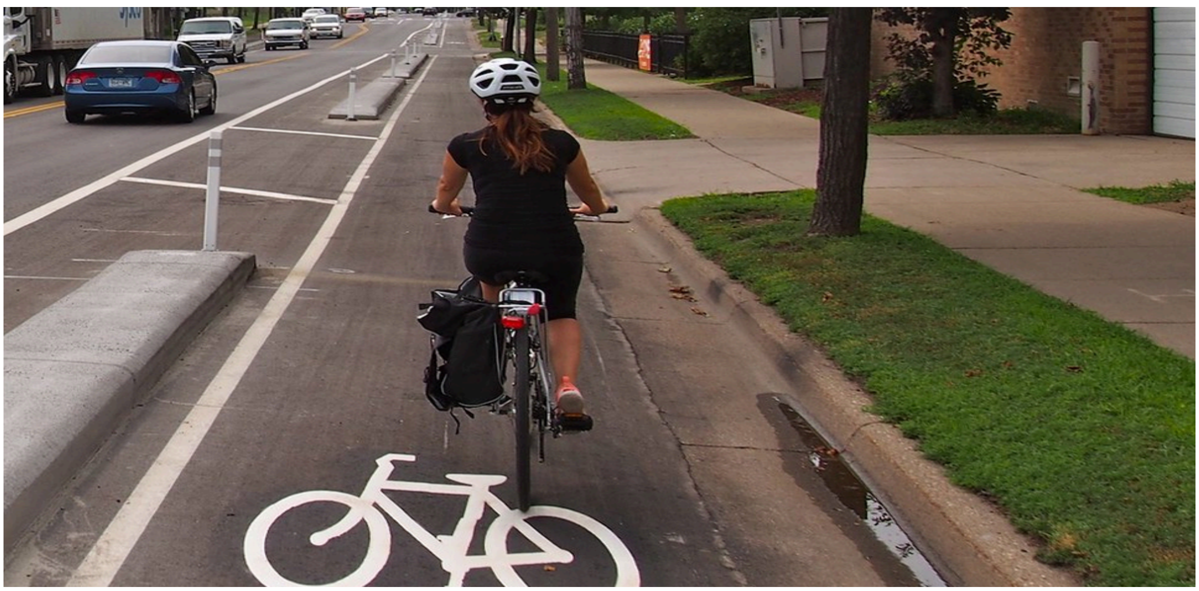
the markings. In high-traffic areas, painted lanes fail to prevent collisions and do not give cyclists the same feeling of safety as physical barriers. This design attempt was unsuccessful at fully solving the safety problem because it relies too much on driver awareness instead of providing structural separation.

Prior Design Attempt #5 - Bike Lane Curbs

“Minneapolis Street Design Guide.” City of Minneapolis Street Design Guide Informs the Planning and Design of All Future Street Projects in Minneapolis, sdg.minneapolismn.gov/design-guidance/bikeways/street-curb-protected-bike-lanes. Accessed 7 Oct. 2025.



Details

Curbed bike lanes are a standard above the average painted bike route without any type of barrier. It includes a small raised concrete curb to distinguish the bike lane from the main road and serves as a barrier for oncoming traffic that prevents cars from swerving into the bike lane. It also requires much less maintenance than other counterpart methods and is more effective in preventing vehicle encroachment.

Analysis

This method requires redesigning the roadway in order to accommodate a sufficient barrier, bike lane, and enough space for the main road. It also requires a minimum of 6 feet in order to account for snow removal. There would also be significantly more city planning due to designing breaks for stormwater drainage and the elimination of parking lanes on the side of roads.